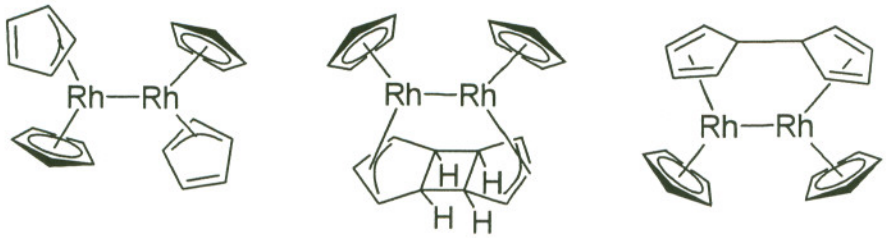
Inorganic Cume, February 6, 2004

1. Although cobaltocene (Cp_2Co) is stable (under N_2) at room temperature, the rhodium and iridium analogs dimerize to give the compounds $M_2C_{20}H_{20}$. Three plausible structures are shown below. Are these 18 electron species? Use the following NMR data to determine the correct structure:
^1H NMR: δ 5.2 (s, 10H), 5.0 (m, 4H), 3.3 (m, 4H), 2.2 (m, 2H) ppm

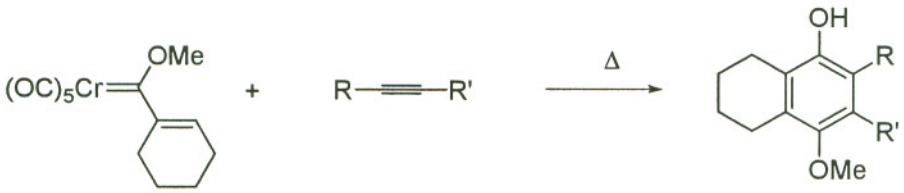


2. Equilibrium constants for the reaction

$$CoBr_2(CO)L_2 \leftrightarrows CoBr_2L_2 + CO$$

for various L are given below.

L	K_d	CO (cm^{-1})	Cone angle (degrees)
PEt_3	(1)	1985	132
$P(n\text{-}Pr)_3$	1.1	1980	135
PEt_2Ph	2.5	1990	135
$PEtPh_2$	24.2	1990	140
PPh_3	750	1995	145

Give the electron count for $CoBr_2(CO)L_2$
Rationalize the trends observed for K_d.

3. Give the mechanism for the Dötz reaction between the following carbene complex and the alkyne indicated. Give details concerning regiochemistry, etc.